Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/01/2006
PI Name:	Puri, Ishwar K. Ph.D.		
Project Title:	Gravitational Effects on Partially Premixed Flames		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Gaseous diffusion flames		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	24061	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	97-HEDS-01
Start Date:	10/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	4
No. of PhD Candidates:	3	**No. of Master' Degrees:**	5
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	17
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:	Note: end date of 9/30/2007 changed. note - end date changed from 11/11/2003; renewed 4 yrs; new grant number		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Aggarwal, Suresh (University of Illinois at Chicago)		
Grant/Contract No.:	NNC04AA06A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Partially premixed flames contain a rich premixed fuel-air mixture in a pocket or stream, and, for complete combustion to occur, they require the transport of oxidizer from an appropriately oxidizer-rich (or fuel-lean) mixture that is present in another pocket or stream. Partial oxidation reactions occur in fuel-rich portions of the mixture, and any remaining unburned fuel and/or intermediate species are consumed in the oxidizer-rich portions. A detailed understanding of the structure of partially premixed flames is important from both practical and scientific considerations. Unwanted fires can originate in a partially–premixed mode when a pyrolyzed or evaporated fuel forms an initial mixture with the ambient air which is fuel rich. Partially premixed flames occur in many applications. We have developed and implemented new measurement techniques for microgravity experiments on open flames (such as analog and digital holographic interferometry, rainbow schleiren, and the use of Fourier optics) through previous NASA–funded research. We have made detailed measurements of normal gravity flames and simulations of partially premixed flames in both normal and microgravity. We have found that (1) heat transport from the nonpremixed to the premixed reaction zones in partially premixed flames, (2) mass transport of radical species among these reaction zones, (3) heat losses (e.g., their effect on pollutant formation) from the reaction zones, and (4) the wide transport region between these zones play an important role in partially–premixed flames. A careful investigation of the interactions between transport, buoyancy, and thermal radiation will increase our understanding of these flames, resulting in better combustor design and fire safety solutions. In general, if the transport zone thickness exceeds 4 mm (as is usually the norm), gravitational effects overwhelm transport effects. Likewise to investigated lifted or propagating flames, the maximum propagation or liftoff height that is uninfluenced by buoyancy effects must be lower than ?1.6 cm for methane–air flames. Clearly, these are restrictive dimensions in which to conduct measurements with suitable clarity and resolution. Furthermore, radiation effects in these flames can be investigated in detail only under microgravity conditions. The flame structure can be modified significantly by changing the level of partial premixing, while keeping other parameters, such as overall stoichiometry, Reynolds number, etc., fixed. Since the flame structure is also influenced by buoyancy (based on the comparison of the diffusive and buoyant time scales), the effect of partial–premixing on the structure can be isolated in a microgravity environment. This has important implications with regard to modifying the flame stability and pollutant emission characteristics of practical combustors. Finally, the investigation of diffusive-thermal instability in the context of partially premixed flames is greatly facilitated in the absence of gravity. The instability, which is related to flame stretch and non-unity Lewis number, is responsible for flame tip opening, and modifying the adiabatic flame temperature and flame speed. It is also directly related to flame liftoff and blowout phenomena. A four-year experimental–numerical investigation is being conducted to investigate these various issues. While premixed and non-premixed microgravity flames have been extensively investigated, the corresponding literature regarding partially premixed flames (PPFs) is sparse. We report the first experimental investigation of burner-stabilized microgravity PPFs. Partially premixed flames with multiple reaction zones are established in microgravity on a Wolfhard-Parker slot burner in the 2.2 second drop tower at the NASA Glenn Research Center. Microgravity measurements include flame imaging, and thermocouple and radiometer data. Detailed simulations are also used to provide further insight into the steady and transient response of these flames to variations in g. The flame topology and interactions between the various reaction zones are strongly influenced by gravity. The flames widen substantially in microgravity. During the transition from normal to microgravity the flame structure experiences a fast change and another relatively slower transient change. The fast response is due to the altered advection as the value of g is reduced, while the slow response is due to the changes in the diffusive fluxes. The radiative heat loss from the flames increases in microgravity. A scaling analysis based on a radiation Damköhler number is able to characterize the radiation heat loss. This investigation has relevance to a variety of industrial applications.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The fire safety lessons learned in space have terrestial applications.
Task Progress:	Partially premixed flames with multiple reaction zones have been established in microgravity on a Wolfhard-Parker slot burner in the 2.2 second drop tower at the NASA Glenn Research Center. Microgravity measurements include flame imaging, and thermocouple and radiometer data. Detailed simulations are also used to provide further insight into the steady and transient response of these flames to variations in g. The flame topology and interactions between the various reaction zones are strongly influenced by gravity. The flames widen substantially in microgravity. During the transition from normal to microgravity the flame structure experiences a fast change and another relatively slower transient change. The fast response is due to the altered advection as the value of g is reduced, while the slow response is due to the changes in the diffusive fluxes. The radiative heat loss from the flames increases in microgravity. A scaling analysis based on a radiation Damköhler number is able to characterize the radiation heat loss. Our experiments are also considering the influence of other diluents on these lifted flames such carbon dioxide and argon. These diluents are the primary components of non-halon gaseous fire extinguishing agents. Both inert species behave uniquely and differently when used as diluents. One reason arises from their differing heat capacities. The diluents also influence the flame by inhibiting the burning intensity, since they increase the third body concentrations and, therefore, influence the combustion chemistry. Different diluents also have differing radiation emission properties, which impact the heat losses from these flames. Our investigation hopes to elucidate these issues in a fundamental sense in the hope of facilitating useful fire safety strategies.
Bibliography Type:	Description: (Last Updated: 12/29/2005)
Articles in Peer-reviewed Journals	Lock, A., Ganguly, R., Puri, I. K., Aggarwal, S. K., and Hegde, U. "Gravity Effects on Partially Premixed Flames: An Experimental-Numerical Investigation" Proceedings of The Combustion Institute, 30 , Jul-2004
Articles in Peer-reviewed Journals	Briones, A., Puri, I. K., and Aggarwal, S. K. "Effect of Pressure on Counterflow H2-Air Partially Premixed Flames" Combustion and Flame , Dec-2004
Articles in Peer-reviewed Journals	Arana, C., Pontoni, M., Sen, S., and Puri, I. K "Field Measurements of Soot Volume Fractions in Laminar Partially Premixed Coflow Ethylene/Air Flames" Combustion and Flame, 138, pp. 362-372 , Aug-2004
Articles in Peer-reviewed Journals	Qin, X., Puri, I. K., Aggarwal, S. K., and Katta, V. R. "Gravity, Radiation and Coflow Effects On Partially Premixed Flames" Physics of Fluids, 16, pp. 2963-2974, 2004 , May-2004

Articles in Peer-reviewed Journals	Qin, X., Choi, C. W., Mukhopadhyay A., Puri, I. K., Aggarwal, S. K., and Katta, V. R. "Triple Flame Propagation and Stabilization in a Laminar Axisymmetric Jet" Combustion Theory and Modelling, 8, pp. 1-22, 2004 , Mar-2004